Supplemental Material for "Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch"

Supplemental Material

Citation numbers follow the numbering of the main article.

S1 Native-Sample Safety and QP Diagnostics

All benchmark violation rates are evaluated at the native 1 s controller instants. A controller sample is counted once when any enforced barrier satisfies $h_j(x_k) < -\eta_j$. Each η_j has the common numerical value 10^{-2} in the native unit of that barrier: MPa for pressure, kJ/kg for separator enthalpy, and MW for power. The test is applied separately to each unrounded constraint value; no minimum is taken across quantities with different units. These tolerances exclude numerical boundary roundoff. Episode-level event probabilities are not combined with these sample rates. Table S1 reports the corresponding count-level S3 diagnostics.

Table S1. Count-level S3 diagnostics for the primary $\Delta g = 0$ validation. Each row aggregates 50 independent 500-sample rollouts. A rejected QP is non-converged, non-finite, or has maximum normalized primal residual above 10^{-6}.

Method	Violating samples	QP rejected	QP intervention
HOCBF, no GP	9798 of 25,000	0 of 25,000	0.00%
RoCBF-SF tune-selected, $\epsilon_\kappa = 0$	0 of 25,000	0 of 25,000	41.32%
Full implemented-margin endpoint, $\epsilon_\kappa = 1$	9798 of 25,000	25,000 of 25,000	0.00%

The tune-selected $\epsilon_\kappa = 0$ row has no rejected QP in S3. The full implemented-margin row rejects every S3 QP, reverts to the upstream proposal, and consequently reproduces the no-GP violation count. This endpoint does not satisfy the robust-QP-feasibility assumption of Theorem 1.

S2 Robustness-Factor Tune/Test Separation

Tune seeds 0–2 each contain 10 episodes of 500 controller samples. Thus, each ϵ_κ setting contains 15,000 tune samples. The predeclared selector is the smallest setting for which both pooled and maximum-per-seed violation rates are below 1% and both pooled and maximum-per-seed QP-rejection rates are below 0.5%. Table S2 gives the complete candidate-level selection data.

Table S2. Tune-seed selection data. Rates are percentages of native controller samples. Wilson upper limits are descriptive Bernoulli-reference values; adjacent samples are serially correlated.

ϵ_κ	Violation count (rate)	Max. seed violation	QP rejection count (rate)	Max. seed rejection	Pass
0	0 (0.000)	0.000	0 (0.000)	0.000	Yes
0.01	0 (0.000)	0.000	0 (0.000)	0.000	Yes

ϵ_κ	Violation count (rate)	Max. seed violation	QP rejection count (rate)	Max. seed rejection	Pass
0.02	0 (0.000)	0.000	0 (0.000)	0.000	Yes
0.05	0 (0.000)	0.000	0 (0.000)	0.000	Yes
0.10	0 (0.000)	0.000	13 (0.087)	0.120	Yes
0.20	4849 (32.327)	37.540	11220 (74.800)	87.660	No
0.50	5663 (37.753)	40.140	14972 (99.813)	99.880	No
1.00	5648 (37.653)	40.000	15000 (100.000)	100.000	No

The selector fixes $\epsilon_\kappa = 0$ before evaluating seeds 3–4. Each held-out seed contains 25,000 samples. Both seeds record zero counted violations under the per-constraint tolerances and zero rejected QPs, giving pooled counts of 0/50,000 for both diagnostics. The descriptive 95% Wilson upper reference for 0/15,000 tune-set violations is 0.0256%; serial correlation prevents interpreting this value as an independent-trial confidence bound. No holdout result is used to change ϵ_κ, and zero observed events are not presented as zero event probability.

S3 GP Residual Definition and Commissioning Data

The benchmark and deployed GP use the same three-dimensional kernel input,

$$x_k^{\mathrm{GP}} = [p_{m,k}, h_{m,k}, N_{e,k}]^\top. \tag{S1}$$

The nominal predictor uses the complete seven-state vector $[r_B, p_m, h_m, N_e, \tau_f, D_{fw}, u_t]^\top$ and all three proposed control commands. The three learning targets are residual rates,

$$y_{j,k+1}^{\mathrm{GP}} = \frac{x_{j,k+1} - \hat{x}_{j,k+1|k}^{\mathrm{nom}}}{T_s}, \qquad j \in \{p_m, h_m, N_e\}. \tag{S2}$$

Kernel distances use the frozen training transform $(x_i - \mu_i^{\mathrm{train}})/s_i^{\mathrm{train}}$, and each residual target is standardized by its training mean and standard deviation. The transforms are not updated online.

The plant commissioning record contains approximately 14 days of 1 Hz historian and commissioning data (about 1.21 million state transitions). Stops, manual operation, communication faults, stale measurements, interlock action, missing values, and identified bad points are removed, leaving approximately 250,000 candidate transitions. Days 1–10 form the training candidate pool; day 11 is a 24 h isolation gap; and days 12–14 form the independent validation pool. Normalization parameters and kernel hyperparameters use only days 1–10. Table S3 specifies the fixed load-stratum quotas used for coverage selection.

Table S3. Fixed plant-GP load strata and quotas. Candidates are separated by at least 60 s and selected by deterministic farthest-point coverage in standardized (p_m, h_m, N_e) space. Quotas are not borrowed between strata. The upper stratum is a coverage bin and does not assert an exact 660 MW steady validation window.

Generator-load stratum (MW)	Training points	Validation points
180–264	100	40

Generator-load stratum (MW)	Training points	Validation points
264–363	100	40
363–462	100	40
462–561	100	40
561–660	100	40
Total	500	200

Three independent exact Matérn-5/2 GPs are trained and frozen after offline replay. The deployed dictionary remains at $N = 500$; online sample growth is zero. Models are reviewed quarterly and after major maintenance or a material coal-source change. The current and previous stable versions are retained for rollback. Exact-GP storage is $\mathcal{O}(qN^2)$ and offline retraining is $\mathcal{O}(qN^3)$ for $q = 3$ outputs. Three float32 Cholesky factors occupy approximately 3 MB; training arrays, coefficients, and kernel parameters occupy approximately 10–20 MB. The deployed JAX process has a larger runtime resident set (approximately 0.5–2 GB), but this does not grow with online operation because the dictionary is frozen.

The controlled sensitivity experiment uses five independent seeds. Each of the nine dictionary-size–corruption settings therefore comprises five seeded fit-and-evaluation runs (45 runs in total). Each run fits three scalar-output GPs, one for each component of $[p_m, h_m, N_e]$, giving 135 scalar GP fits. It then evaluates 200 held-out transitions and one 500-sample S3 closed-loop rollout. Corruption replaces 5% or 10% of selected training targets by signed three-training-standard-deviation offsets; it does not represent a measured plant bad-point rate. Table S4 reports the held-out and closed-loop sensitivity results.

Table S4. GP quantity/quality sensitivity. NRMSE and interval coverage are averaged over outputs and seeds. Closed-loop columns aggregate 2500 native controller samples per cell.

N	Target corruption	Held-out NRMSE	95% interval coverage	Violation, $\epsilon_\kappa = 0$	QP rejection, $\epsilon_\kappa = 0$	QP intervention
100	0%	0.00090	100.00%	0.00%	0.00%	41.80%
250	0%	0.00055	100.00%	0.00%	0.00%	41.76%
500	0%	0.00046	100.00%	0.00%	0.00%	41.76%
100	5%	0.54291	96.03%	20.24%	6.76%	24.00%
250	5%	0.41965	97.13%	5.76%	4.04%	37.12%
500	5%	0.06085	99.90%	0.00%	0.00%	41.92%
100	10%	0.66568	96.37%	15.12%	8.60%	27.16%
250	10%	0.44662	97.03%	13.16%	5.88%	33.20%
500	10%	0.08745	99.80%	0.00%	0.00%	41.32%

The corrupted fits are deliberately evaluated without the pre-enable data-quality gate. The 100- and 250-point dictionaries show increased violations and QP rejection, while the 500-point dictionary retains zero counted violations under the per-constraint tolerances and no rejection under both bounded fault-injection levels. The comparison therefore tests the joint effects of dictionary coverage and target quality; it does not imply that every corrupted fit fails or that the 500-point model is robust to arbitrary bad data. In deployment, held-out residual, coverage, and QP-feasibility checks are all required before a model receives write authority.

S4 Coverage Monitoring and Fallback State Machine

The online input-coverage statistic and standardized innovation are

$$z_{i,k} = \frac{x_{i,k}^{\mathrm{GP}} - \mu_{x_i}^{\mathrm{train}}}{s_{x_i}^{\mathrm{train}}},$$
$$e_{j,k+1} = \frac{|r_{j,k+1} - \mu_j(x_k^{\mathrm{GP}})|}{\sqrt{\sigma_j^2(x_k^{\mathrm{GP}}) + \sigma_{n,j}^2}}, \quad r_{j,k+1} = \frac{x_{j,k+1} - \hat{x}_{j,k+1|k}^{\mathrm{nom}}}{T_s}. \tag{S3}$$

Here $r_{j,k+1}$, μ_j, σ_j, and $\sigma_{n,j}$ all have residual-rate units. Because the deployed sampling period is $T_s = 1$ s, this notation correction does not change the reported numerical thresholds. Table S5 lists the monitor triggers and associated supervisory actions.

Table S5. GP coverage and degradation thresholds used by the deployed supervisor.

Monitor	Trigger	Action
Input coverage	Any $\lvert z_i \rvert > 3$ or outside the training 0.5th–99.5th percentile	Enter full-row nominal-HOCBF mode.
Posterior uncertainty	Above the validation-set 99th percentile	Withdraw GP mean correction.
Standardized innovation	Any $e_j > 3$ for three cycles	Withdraw GP mean correction and alarm.
Severe innovation	Any $e_j > 5$ once	Immediate live-DCS bypass.
Persistent OOD	More than 1% of cycles in one hour	Create an offline recalibration task.
Recovery	Ten consecutive cycles with valid input, innovation, and variance checks	Permit GP-mode recovery subject to controller authorization.

Moderate coverage loss follows

$$\begin{gathered}\text{GP-RoCBF-SF} \to \text{nominal HOCBF} \quad \text{(for at most 10 cycles)},\\ \text{nominal HOCBF} \to \text{GP-RoCBF-SF} \quad \text{(after 10 healthy cycles)},\\ \text{nominal HOCBF} \to \text{BYPASS} \to \text{live DCS} \quad \text{(otherwise)}.\end{gathered} \tag{S4}$$

Invalid measurements or communication, non-finite/model-load faults, nominal-QP or actuator-constraint infeasibility, a direct physical-bound exceedance, pressure-low physical margin below 2.0 MPa, or a QP/task timeout bypasses the bounded degraded stage immediately. The nominal-HOCBF interval has no GP coverage claim.

An initially infeasible QP permits one reduced-QP retry only for the whitelisted pressure_low row. Let the physical command deviation be $d_u = S_u v$, with normalized $v \in [-1,1]^3$, transformed row $\widetilde{G}_i = G_i S_u$, and authority measure $a_i = \| \widetilde{G}_i \|_2$. Row removal requires

$$i = \text{pressure\_low}, \quad a_i < 0.01, \quad h_i < -10^{-6}, \quad PM - PLO \geq 2.0\ \text{MPa}, \tag{S5}$$

with valid measurements and no active protection or interlock. The authority norm is evaluated after actuator normalization but before unit-row scaling. No actuator box/rate row or second CBF row can be removed, and only one retry is permitted. A successful reduced QP is reported as optimal_recovered; it is outside the full-row certificate.

The deployed task deadline is 1000 ms, equal to the controller period. The field QP timeout is 200 ms. A timeout, non-finite result, or failed retry returns the current live DCS command rather than the previous algorithm command. Three consecutive solve failures, communication loss beyond 3 s, a process restart, or a model-load fault latches bypass. Re-entry requires valid measurements, heartbeats, and dry-run QPs for ten cycles, DCS automatic mode, inactive protection/interlocks, operator reset acknowledgement, and the active Kubernetes lease. A 10 s linear blend is then used for bumpless transfer, provided the QP and DCS commands differ by no more than 2% of actuator range and both lie in the current feasible set.

Kubernetes uses a 3 s Lease renewed every 1 s. Only the lease holder may assert the local write permit; standby Pods connect to Modbus TCP read-only. The DCS remains the final write-permission owner and requires enable, automatic mode, no manual mode, valid echoed heartbeat with age no greater than 2 s, active lease ownership, and a fresh command sequence. Loss of any condition selects the live coordinated-control command.

S5 Plant Historian and Controller-Export Evidence

The public production-evidence index separates historian context from controller execution. MW01–MW03 are post-retrofit low-to-mid-load historian windows used only to position measured load and main-steam pressure in the operating-envelope figure; they do not supply controller-internal QP fields. MW04–MW06 are confirmed two-hour plant-controller exports covering 480.7–629.7 MW. Each contains 1440 records at 5 s export spacing from a controller executing at 1 s, giving 4320 exported records rather than 4320 controller cycles. These exports directly support the timestamps, operating mode, QP/recovery status, logged margins, saturation flags, and timing statistics reported below. GP lifecycle, Kubernetes lease ownership, heartbeat/sequence checks, and DCS write-permission rules are deployment-configuration specifications; the exports do not expose every internal handshake transition.

The confirmed controller/historian mappings used in the revision are $NE \leftrightarrow$ 20CQTP_MW, $PM \leftrightarrow$ DCS2_20HAG10CP101, and $HM \leftrightarrow$ DCS2_SEPARATOUT_ENTH. The independent tag DCS2_MAIN_PRESS is main-steam pressure. The exported PST_HI value is a main-steam-pressure replay bound, not the separator-pressure upper limit.

The cohort analysis uses 300 s historian records from 2024–2025 and 25 June–5 July 2026. Two-hour windows contain 24 consecutive records and advance every 30 min. All 24 load samples must lie in 195–700 MW; this upper value is a permissive data-quality screen, not a reported operating endpoint. Pressure, pressure setpoint, and fuel flow must be finite, pressure channels must lie in 5–35 MPa, and fuel flow must be positive. One-to-one matching without replacement minimizes a standardized sum of squared differences in mean load, 24-point load-profile RMSE, load range, and mean fuel/load ratio. The respective calipers are 10, 30, and 40 MW and 0.08. The complete derived pair table and hashes of the 24 monthly pre-retrofit query files and the post-retrofit query file are included in the reproducibility package. Table S6 summarizes the recomputed matched cohort and its descriptive intervals.

Table S6. Recomputed load-matched historian cohort. Windows overlap because they advance every 30 min; the post-day cluster-bootstrap intervals are therefore descriptive and are not randomized-treatment confidence intervals.

Quantity	Pre-retrofit	Post-retrofit	Difference or interval
Loaded candidate windows	14,675	517	512 one-to-one pairs
Unique calendar days in pairs	193	11	–
Pressure-residual standard deviation, median (MPa)	0.523	0.502	3.93% reduction
Absolute pressure-residual p95, median (MPa)	1.082	1.085	0.30% increase
Pair fraction with lower standard deviation	–		55.27%
Pair fraction with lower absolute-error p95	–		59.18%
Post-day cluster bootstrap, standard-deviation reduction	10,000 resamples		95% interval [-13.0,14.7]%
Post-day cluster bootstrap, p95 reduction	10,000 resamples		95% interval [-9.4,13.1]%

Median pair-quality diagnostics are 1.32 MW mean-load difference, 15.37 MW load-profile RMSE, 5.24 MW load-range difference, and 0.0115 fuel/load difference. The broad cluster-bootstrap intervals in Table S6 show that the cohort supports operating-context consistency but not a precise population effect estimate. The native 5 s pair in the main article provides the higher-resolution response diagnostic; the controller exports separately establish online filter execution. Table S7 then summarizes the separate high-load controller-export evidence.

Table S7. High-load controller-export evidence. Direct margins are minima over all 1440 records in each window. "None" in the status column denotes no reduced-QP recovery and no saturation flag.

Window	Load (MW)	Sep.-pressure low margin (MPa)	Main-pressure high margin (MPa)	Nearest enthalpy margin (kJ/kg)	QP/recovery record	QP/task p95 (ms)
MW04	563.0–621.2	2.217	2.173	62.284	1397 optimal; 43 recovered; 1 fuel-sat.	4.195 / 26.238
MW05	547.9–629.7	3.438	1.483	20.354	1440 optimal; none	4.074 / 26.139
MW06	480.7–619.3	2.705	0.463	0.359	1440 optimal; none	4.062 / 26.159

All 43 MW04 recovery records have FB_RSN=weak_authority_constraint_drop, ACT_CSTR=pressure_low, and negative MG_PLO. The diagnostic row ranges from -0.03266 to -0.00561, while the direct physical margin $PM - PLO$ ranges from 2.21654 to 2.36306 MPa and the other five exported HOCBF margins remain positive. MW05–MW06 contain no reduced-QP record. Across MW04–MW06, QP time has median/p95/maximum values 3.557/4.095/6.234 ms, and total task

time has corresponding values 24.564/26.183/28.665 ms.

The plant value $\epsilon_{\kappa} = 0.1$ is fixed after training-block replay and passes the one-pass time-isolated validation gate without retuning; it is not copied from the benchmark S3 selection. Record-level plant replay remains a restricted enterprise asset and is not inferred from the high-load controller exports. Retained plant work-scope and controller-configuration records document that the original PID gains, output limits, anti-windup, sliding-pressure curve, coordinated-control logic, and fuel/feedwater/turbine-valve base loops were unchanged; RoCBF-SF was added downstream with enable, bypass, heartbeat, command-quality, and DCS-selection logic. The same work-scope records document instrument calibration, actuator inspection, combustion-equipment maintenance, and heat-surface cleaning during the outage. The record summary therefore narrows the control-system change to the proposed integration but does not remove concurrent maintenance as an alternative contributor, so the historian before/after comparison remains observational. The controller exports separately establish that the proposed filter executed after restart.

The public reproducibility package contains source code, simulation results, derived historian metrics, bounded controller-export excerpts, field maps, and source/public hashes. Complete raw historian and original controller exports are enterprise assets. Qualified researchers may request restricted access from the corresponding author, subject to data-owner approval and the applicable data-use conditions.

S6 Computation and Sampled-Data Boundary

The NMPC reference uses a warm-started SLSQP solve with horizon $N = 5$, output weights diag(1,0.001,0.01), input weights diag(0.01,0.01,0.01), normalized deviation-input bounds $[-1,1]^3$, six prediction constraints, additive disturbance correction with gain 0.5, maxiter=50, and ftol= 10^{-4}. Its convergence, residual, and computation-time statistics are regenerated with the actuator-augmented benchmark and reported from that frozen result set. These values describe the implemented reference rather than an optimized industrial MPC package.

Across the seven confirmatory conditions, pooled research-pipeline time per controller update ranges from 3.19 to 6.71 ms for the tune-selected RoCBF-SF row and from 3.75 to 20.82 ms for NMPC. NMPC records no solver failure and 0/175,000 counted state-limit violations under the per-constraint tolerances in this implementation. These benchmark wall-clock measurements are distinct from the deployed-CPU evidence: the confirmed controller exports report a maximum QP time of 6.234 ms and a maximum total task time of 28.665 ms against the 1000 ms task deadline.

The theorem is continuous-time. The controller applies zero-order hold with $T_s = 1$ s because 1 s is the deployed coordinated-control task period; the safety filter therefore evaluates each upstream command update before the command-selection stage. This implementation choice does not assign a 1 s time constant to the slower boiler thermal dynamics. The field exports retain one record every 5 s, so their spacing is a logging interval rather than the internal control period. Field task timing establishes computational fit within the 1 s period but does not establish inter-sample forward invariance. Theorem 1 therefore is not presented as a sampled-data certificate; deriving such a certificate is retained as future work.

S7 Proofs of Theoretical Results

S7.1 Residual Cross-Term Bound

Lemma S1 (Residual cross-term bound). Let $\delta_i = \psi_i - \widehat{\psi}_i^0$ with $\delta_0 \equiv 0$. On the simultaneous residual event in Assumption 2 of the main article, suppose a valid pointwise derivative bound $\| \nabla_x \delta_{i-1}(x) \|_2 \leq G_\delta^{(i-1)}(x)$ is available on $\mathcal{X}_{\mathrm{op}}$. Then

$$|L_{\Delta \hat{f}} \delta_{i-1}(x)| \leq G_\delta^{(i-1)}(x) \beta \| \sigma_{\mathrm{GP}}(x) \|_2. \tag{S6}$$

Proof. By Cauchy–Schwarz,

$$|L_{\Delta \hat{f}} \delta_{i-1}| = |\nabla_x \delta_{i-1}^{\top} \Delta \hat{f}| \leq \| \nabla_x \delta_{i-1} \|_2 \| \Delta \hat{f} \|_2. \tag{S7}$$

The simultaneous residual event gives $\| \Delta \hat{f}(x) \|_2 \leq \beta \| \sigma_{\mathrm{GP}}(x) \|_2$ over the modeled outputs, which yields Eq. (S6). Smoothness on a compact operating set ensures that finite global derivative bounds exist when the required derivatives are bounded, but it does not identify them with $\| \nabla \widehat{\psi}_{i-1}^0 \|_2$. □

The online code uses $\| \nabla \widehat{\psi}_{i-1}^0 \|_2$ as a commissioning proxy for $G_\delta^{(i-1)}$. Theorem 1 applies to that implementation only when Assumption 3 of the main article independently verifies that the resulting total margin dominates the true top-level HOCBF residual on $\mathcal{X}_{\mathrm{op}}$.

S7.2 Full Proof of Theorem 1

Proof of Theorem 1. **Step 1 (Simultaneous residual event).** Assumption 2 of the main article states that $|\Delta f_j(x) - \mu_{\mathrm{GP},j}(x)| \leq \beta \sigma_{\mathrm{GP},j}(x)$ holds simultaneously for the $q = 3$ modeled residual outputs and all $x \in \mathcal{X}_{\mathrm{op}}$ with probability at least $1 - \delta$. Condition on this event. The commissioning multiplier used in the numerical studies does not establish this event by itself.

Step 2 (Compositional bound). We show $|\delta_i(x)| \leq \sigma_i(x)$ by induction on i.

Base case ($i = 1$): $\delta_1 = \sum_j (\partial h / \partial x_j) \Delta \hat{f}_j$. By Step 1 and triangle inequality, $|\delta_1| \leq \beta \sum_j |\partial h / \partial x_j| \sigma_{\mathrm{GP},j} = \sigma_1$.
Inductive step ($i \geq 2$): Assume $|\delta_{i-1}| \leq \sigma_{i-1}$. The residual recursion in the main article gives $|\delta_i| \leq |L_{\Delta \hat{f}} \widehat{\psi}_{i-1}^0| + (|L_{\hat{f}}|_{\mathrm{op}} + k_i) |\delta_{i-1}| + |L_{\Delta \hat{f}} \delta_{i-1}|$. The first term is bounded as in the base case. The second term is bounded by the induction hypothesis. Lemma S1 bounds the third term when a valid $G_\delta^{(i-1)}$ is available. Assumption 3 requires the operator-norm and cross-term quantities used for the certificate to dominate these terms. Summing gives $|\delta_i| \leq \sigma_i$.

The implementation proxy is not substituted for $G_\delta^{(i-1)}$ in this proof unless Assumption 3 validates that substitution.

Step 3 (Total perturbation). Combining the inductive bounds from Step 2 with the coupling-coefficient bound in Eq. (S8) yields $|\Delta_\psi(x, v)| \le \sigma_m(x) + \sum_{j=1}^{m-1} c_j\,\sigma_j(x) + \sigma_{\mathrm{ctrl}}(x) = \sigma_{\mathrm{total}}(x)$ for every admissible deviation input v. For $\epsilon_\kappa = 1$, the applied margin is $\sigma_{\mathrm{total}}(x)$.

Step 4 (Forward invariance). For v^* satisfying $A(x)v^* \le b(x) - \sigma_{\mathrm{total}}(x)$, the definition $\widehat{\psi}_m^0(x, v) = b(x) - A(x)v$ gives $\widehat{\psi}_m^0(x, v^*) \ge \sigma_{\mathrm{total}}(x)$. Hence $\psi_m^{\mathrm{true}}(x, v^*) = \widehat{\psi}_m^0(x, v^*) + \Delta_\psi(x, v^*) \ge \sigma_{\mathrm{total}}(x) - |\Delta_\psi(x, v^*)| \ge 0$. Because $x(0) \in \mathcal{C}$, all lower-order recursive conditions in the main article's definition of $\mathcal{C}$ are satisfied initially. The HOCBF forward-invariance theorem cited there therefore renders $\mathcal{C}$ forward invariant. □

S7.3 Coupling Coefficient Gap Bound

The coupling coefficient gap is bounded by:

$$\sigma_{\mathrm{ctrl}}(x) = \beta v_{\max} \sum_j \left\| \frac{\partial (L_{g_v} L_{\hat{f}}^{m-1} h)}{\partial x_j} \right\|_2 \sigma_{\mathrm{GP},j}(x) \tag{S8}$$

where $g_v = g_0 S_u$ and $v_{\max} = \sup_{v \in [-1,1]^3} \| v \|_2 = \sqrt{3}$. This term is evaluated separately for each barrier row and included in that row's total margin; Theorem 1 does not assume a fixed ratio between σ_{ctrl} and σ_{total}.